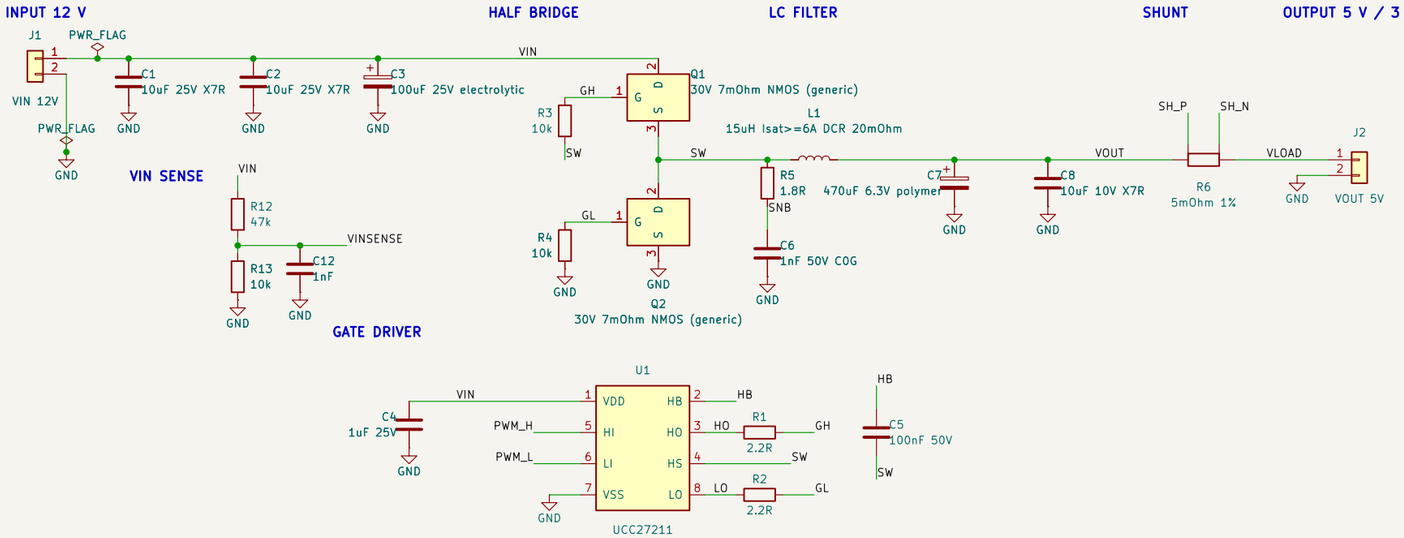
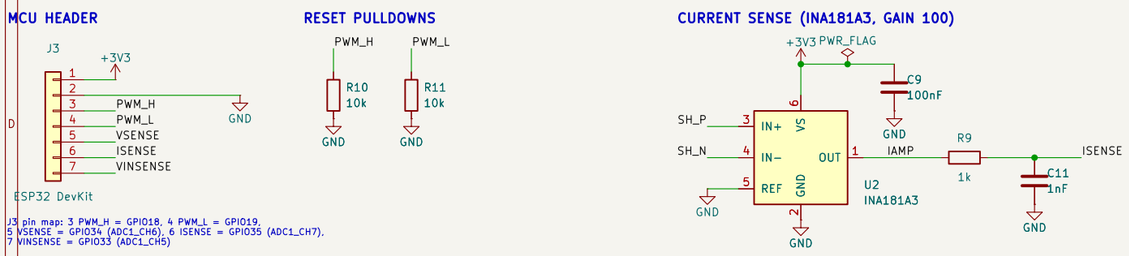


DESIGN NOTES (simulated design, nothing built)
Buck: VIN 12 V -> VOUT 5 V / 3 A, ideal duty D = 5 / 12 = 0.417
Q1 / Q2: generic 30 V NMOS, pin order G = 1, D = 2, S = 3 (DPAK TO-252 pad order)
U1 UCC27211: VDD fed from VIN, C4 1 uF decoupling; R1 / R2 2.2 Ω gate resistors; R3 / R4 10k pulldowns
Snubber SW -> R5 1.8 Ω -> SNB -> C6 1 nF COG -> GND
R6 shunt: 5 mΩ 1 %, 4-terminal Kelvin (force pins 1 / 4, sense pins 2 / 3)
  V = 3 A x 5 mΩ = 15 mV; P = (3 A)² x 5 mΩ = 45 mW
U2 INA181A3 (gain 100 V/V), REF = GND, VS = +3V3; ISENSE = 100 x 15 mV = 1.5 V at 3 A
  R9 1k + C11 1 nF anti-alias: fc = 1 / (2π x 1k x 1 nF) = 159 kHz
VSENSE divider: 5 V x 6.8k / (10k + 6.8k) = 2.02 V; R7 || R8 = 4.05k with C10 1 nF: fc = 39 kHz
R10 / R11 10k pulldowns keep PWM_H / PWM_L low (FETs off) while the ESP32 is in reset
VINSENSE divider: 12 V x 10k / (47k + 10k) = 2.11 V; R12 || R13 = 8.25k with C12 1 nF: fc = 19 kHz
J3 ESP32 DevKit: 1 +3V3, 2 GND, 3 PWM_H (GPIO18), 4 PWM_L (GPIO19),
  5 VSENSE (GPIO34, ADC1_CH6), 6 ISENSE (GPIO35, ADC1_CH7), 7 VINSENSE (GPIO33, ADC1_CH5)



J3 pin map: 3 PWM_H = GPIO18, 4 PWM_L = GPIO19,
5 VSENSE = GPIO34 (ADC1_CH6), 6 ISENSE = GPIO35 (ADC1_CH7),
7 VINSENSE = GPIO33 (ADC1_CH5)

U1 UCC27211, U2 INA181A3 (gain 100), 5 mOhm Kelvin shunt, ESP32 DevKit header
Simulated design only: nothing built in hardware

Shon Zinger

Sheet: /
File: sync_buck.kicad_sch

Title: 02 — Sync buck 12 V to 5 V / 3 A (simulated)

Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1